

Who is this meeting with?

The one question every pre-meeting brief depends on — and the six rungs the system climbs to answer it, from a free lookup in your own directory to a scored search across two public registries. With the test results that shaped each rule.

Repo outlook-calendar-poc **Branch** feat-outside-bis-physician-pre-notes

Tests 104 passing **Verified** 1 September 2026, live NPPES + CMS + Supabase

01 The question, and the order it is asked in

A rep opens Outlook and sees eight meetings. Three are with physicians, one is a pipeline review, and one says nothing more than “Dr Khan”. Every brief the system produces — the email, the notes inside the meeting, the reminder — rests entirely on identifying the right person, and identifying the wrong one is worse than identifying nobody: it hands a rep another physician’s procedure volumes under the name they expected to see.

So the answer is built as a **ladder**. Each rung is a stronger claim than the one below it and costs more to check, so the cheapest sufficient answer always wins and nothing external is touched until everything free has been exhausted. The order is not a preference; it is the confidence ranking.

RUNG	WHAT IS CHECKED	COST	IF IT ANSWERS
0	An NPI written on the meeting	free, 0 ms	resolved • 100%
1	An attendee's email , exactly, in the master	free, 0 ms	resolved
2	The physician the rep already confirmed for this meeting	free, 0 ms	resolved
3	The gate: does the meeting say “Dr” / “Doctor” ?	free, 0 ms	opens rung 4, or stops here
4	That name in the master (21,274 physicians, in memory)	free, 0 ms	one hit, a shortlist, or half a name
5	Public registries — NPPES by name, then CMS by NPI	network	scored candidates, best first

Rungs 0–4 never leave the building. Only rung 5 makes a network call, and only for a meeting the master could not answer.

A meeting appears, or changes

Outlook · read by the ingest tick every 300 s , or when the rep opens the day view

**Rung 0 — is an NPI written on it?**

title, description or location · validated against the NPI check digit, so a phone number cannot pass

In the master

the standard brief

Not in the master

straight to the registries *by NPI* — nothing to disambiguate

no NPI on the meeting

Rung 1 — is an attendee's email in the master?

exact match only · the organiser and meeting rooms are never matched

Yes → every match

two physician attendees produce two briefs, in one email

No

an email cannot be searched outside: neither registry indexes one

**Rung 2 — has the rep already confirmed someone here?**

their pick is stored per meeting and outranks every guess below

no stored decision

Rung 3 — the gate: “Dr” or “Doctor”?

in the title, or in an attendee's display name · whole words, so “Drainage” and “Andrew” do not open it

Gate shut → a normal meeting

no name is read, no lookup runs, nothing is spent. The panel says so and offers a search box.

gate open

Rung 4 — that name in the master

the title's name, or an unmatched attendee's display name · honorific and credentials stripped

Exactly one

resolved — the standard brief

Several

a shortlist, ≤ 5 , facility and city first

Half a name

“Khan” → 62 in the master · the rep is asked for the missing half

None

on to the registries



RUNG 5 · OUTSIDE THE MASTER — THE ONLY NETWORK PATH

NPPES, by name

either half of a name works · up to 20 fetched, then scored — asking for five could miss the right person entirely



Score every candidate against the meeting

first and last name, city, state, and whatever the meeting itself mentions — taxonomy, practice address, ZIP, phone

≥ 70% and clear of the runner-up

shown as the answer, with its reasons

60–69%, or a tie

offered as options, each with its percentage

Under 60%

not shown at all — the count is stated and the rep is asked for one more detail



CMS, by the NPI that came back

Medicare claims per year, 2024 and 2023 · CPT lines, services, patients, allowed amounts



Cross-confirm

do NPPES and CMS agree on surname, city and state? agreement is stated and adds confidence



The brief — one renderer, four places

the app panel · an email the rep can send themselves · the meeting's own body in Outlook · the reminder, 30 minutes before

Every box that stops the flow says why. A rung that cannot answer hands over; a rung that *could not run* — a registry that did not resolve — says that instead, because “we could not ask” and “there is nobody” are different answers and a rep acts differently on each.

Step by step, with what actually happened

Each rung below was run against the live system on 1 September 2026 — the real Supabase directory, the real NPPES API, the real CMS datasets. The examples are the outputs, not illustrations.

0

NPI

An NPI written on the meeting

Ten digits a rep pasted in is the strongest identifier there is: no name to parse, nothing to disambiguate, and no reason to require “Dr”. It is checked before everything else — and validated against the **NPI check digit** (a Luhn sum over the number prefixed with 80840), because a meeting body is full of ten-digit numbers.

```
"Review NPI 1003000126"
```

```
rung 0 → 1003000126 is on the meeting, not in the master
NPPES → ARDALAN ENKESHAFFI · Hospitalist · BETHESDA, MD
CMS 2024 → 384 services · 341 patients · $44,804 allowed · 7 codes
CMS 2023 → 802 services · 556 patients · $77,382 allowed · 7 codes
brief → 100%, "the NPI was written on the meeting itself"
        confirmed by NPPES and CMS (surname, city, state agree)
```

Rejected by the same rule: 7135550100 (a phone number), 1234567890, and any nine-digit id — all pass a length check and fail the check digit, which is exactly why the check digit is used.

1

EMAIL

An attendee's email, exactly

The invite's own address list, matched exactly against the master. Every match counts, so a meeting with two physician attendees produces two briefs in one email. The **organiser is never matched** — that is the rep — and neither are meeting rooms.

```
"Quarterly review" · attendee aaron.baas@avera.org
```

```
rung 1 → matched exactly: Aaron P Baas (NPI 1114144375)
brief → General Surgery · Avera Queen of Peace, Mitchell SD
        email flagged "⚠️ unverified — from the BIS master,
        never confirmed with the physician"
        ✓ practice location matches the NPPES registry (MITCHELL, SD)
        verified mobile + LinkedIn · contactability 92%
        Colonoscopy 270 (CPT 45378, 45380, 45384, 45385)
        therapeutic adoption 74% · snare used 16%
        Lumendi account: active — DiLumen EZ since 2025-03-15
```

When the email is *not* in the master, the flow moves to the name — not because a name is better, but because **neither public registry can be searched by email**. NPPES has no email field at all.

2

PICKED

The physician the rep already confirmed

A shortlist is a question; this is the answer to it. Asking again every few minutes is worse than useless — it re-lists candidates the rep has already ruled out. The pick is stored per meeting, with who made it and when, and it outranks every automatic guess below it.

It also fixed a bug that used to destroy it: the ingest tick writes the meeting's physician on every sync, and for a meeting with no matching attendee email that value was `null` — written straight over the rep's choice. A confirmed choice now outranks any automatic match, and an automatic match never overwrites a known link with nothing.

3

GATE

The gate — the meeting must call them a doctor

Without a gate, “Pipeline review” and “1:1 with Sam” become physician lookups. With it, only a meeting that says **“Dr”, “Dr.”, “Drs” or “Doctor”** opens the name path. Whole words only, so “Drainage” and “Andrew” do not qualify.

The title is the usual place but not the only one: an invite whose attendee reads *“Dr Nicholas Shaheen <nshaheen@med.unc.edu>”* has said it just as plainly, and an address that fails to match the master is precisely when the name path is needed. Requiring the honorific in the title alone left those meetings with nothing — found while testing, and fixed.

```
"Endoscopy sync" · attendee Dr Nicholas Shaheen <nshaheen@med.unc.edu>
```

```
rung 1 → the address is not in the master
rung 3 → gate opens on the ATTENDEE name, not the title
rung 5 → NICHOLAS J SHAHEEN, NPI 1467521757 — 100%
CMS    → 2024: 292 services · 232 patients · $37,098
        2023: 290 services · 212 patients · $38,505
        43239 biopsy (office 76, facility 49) · 43229 polyp
        destruction 27 · 43249 balloon dilation 26
```

4

NAME

That name, in your own directory

The 21,274-physician master is already in memory, so this is free and complete where a paged search is neither. A stored middle initial does not block a match — “Aaron Baas” finds “Aaron P Baas” — and four outcomes are possible.

"Case obs with Dr Aaron Baas"	→ one hit → the standard brief
"Meeting with Dr Abdul Khan"	→ 3 hits → shortlist, rep picks
"Meeting with Dr Khan"	→ half a name → 62 hits + an ask
"Meeting with Dr Charlotte Aagaard"	→ none → the registries

Half a name is kept, not dropped. “Khan” alone identifies nobody — it is 62 physicians in the master — but the old rule discarded it and left the rep with an empty panel and no explanation. Now the master is searched by surname and the notes carry a specific ask: *“the first name is missing. ‘Khan’ alone matches 62 physicians.”*

Which half is missing is answered by the master itself, not guessed: “Khan” ends sixty-odd stored names and starts almost none, so it is a surname and the **first** name is what is needed. “Geoffrey” is the reverse. When the distribution cannot tell, the ask is for the full name rather than for the half the rep already gave.

Read once, not twice. Half names go through the same reader as whole ones, with the minimum word count relaxed to one. Re-implementing the parse is what once turned “*Dr rounds*” into a physician called Rounds — the meeting words, place words, capitalisation rule and organiser exclusion all live in one place, and they all still apply.

5

OUTSIDE

The public registries

Two sources, each doing only what it is good at. **NPPES** turns a name into an NPI; **CMS** turns that NPI into what the physician actually does. The order is deliberate — CMS’s table has one row per provider *per code*, so a name query there returns hundreds of rows for one person and dozens of same-named strangers.

Adding a third website is a module and one line: a source declares its id, its human page, whether it can search by name, and returns candidates in the shape the rest of the system already renders. Nothing else changes.

04

How sure are we? The number, and why it exists

Nine people share the surname Aagaard. Listing all nine at 55% teaches a rep to click through noise; presenting one of them as the answer teaches them to trust a coin toss. So every candidate is scored against **what the meeting itself said**, the reasons are kept alongside the number, and two thresholds decide what is shown.

SIGNAL	Δ	WHY IT IS WORTH THAT
A registry listed them at all	+30	a floor, not evidence
Last name matches	+25	the field the registry indexes
First name matches exactly	+30	the single strongest name signal
First name is a prefix / an initial	+15 / +10	an initial is shared by dozens
First name differs	−25	a different person with the same surname
State / city matches the meeting	+15 / +15	geography separates same-named providers
Primary taxonomy matches (exact / contained)	+25 / +20	only one of nine Aagaards is a dentist
Primary taxonomy contradicts the meeting	−15	a listed taxonomy can still be phrased differently
Practice address / ZIP / phone matches	+20 each	the fields a rep can copy from the registry page
The only candidate any source returned	+10	nothing to be confused with
A second source confirms the identity	+10	two independent registries agreeing

Capped at 100. The reasons travel with the number, so “85%” is shown as “85% — last name matches, first name matches, state matches the meeting”.

≥ 70% shown as the answer

...and only when it is **at least 10 points clear** of the runner-up. Its brief is assembled and rendered without the rep clicking anything.

60–69% offered

Listed as options, each with its own percentage, best first, at most five. A tie above 70% lands here too: if two candidates cannot be told apart, neither is the answer.

under 60% withheld

Not shown at all — but **counted out loud**. “The registries returned 9 people named ‘Aagaard’, but nothing in this meeting says which one.”

What the meeting mentions is what disambiguates

Rather than trying to parse a taxonomy or an address out of a free-text title — a guessing game — the scorer looks for *each candidate’s own values inside the meeting text*. “Does this meeting say ‘Dentist?’” has a definite answer, and it is exactly how a rep disambiguates by hand.

THE MEETING SAYS...	CHARLOTTE SUE AAGAARD BEHAVIOR TECHNICIAN • SEATTLE WA	JEFFREY LYNN AAGAARD DENTIST • WEST DES MOINES IA
Dr Aagaard	55% withheld	55% withheld
Dr Aagaard (Dentist)	55% withheld	75% shown
Dr Aagaard at 4800 Sand Point Way NE	75% shown	55% withheld
Dr Charlotte Aagaard	95% shown	40% withheld

Real NPPES records, real scores. One line added to a meeting title moves the answer from “nothing shown” to “exact”. These are *identity* scores — whether a brief is then produced is a separate question, answered in §05: Charlotte Aagaard is identified at 95% and briefed at none, because a behaviour technician is not a doctor.

05 Who gets a brief at all

A national registry lists whoever holds an NPI, and an NPI is not a medical degree. The nine people called AAGAARD include four physicians, a dentist, a counsellor, a social worker, a behaviour technician and a physical therapy assistant. A pre-meeting brief for the social worker is worse than no brief: it spends the rep’s two minutes on the wrong person and implies the app checked who they were.

So a brief is produced for **physicians, dentists and podiatrists**. Everybody else is *named, not briefed* — the panel says who the registry says they are, and links the page that proves it.

Decided on the code, not the words. NPPES returns each provider’s NUCC taxonomy code (207RC0001X), and its first two digits are the official Provider Grouping — a structured, stable fact. Classifying by description would call a **Physician Assistant** a physician, because the word is right there.

NUCC	GROUPING	EXAMPLE, AS THE REGISTRY WRITES IT	BRIEF?
20	Allopathic & Osteopathic Physicians	207RC0001X Internal Medicine, Cardiac EP	yes
21	Podiatric Medicine & Surgery	213E00000X Podiatrist	yes
12	Dental Providers — <i>only</i> 1223...	122300000X Dentist	yes
12	...the rest of the dental grouping	124Q00000X Dental Hygienist	no
10	Behavioral Health & Social Service	104100000X Social Worker · 101YM0800X Counselor · 106S00000X Behavior Technician	no
17	Other Service Providers	171M00000X Case Manager/Care Coordinator	no
36	Physician Assistants & Advanced Practice Nursing	363A00000X Physician Assistant	no
16 · 18 · 22 · 23 · 24 · 25 · 11 · 15	Nursing · Pharmacy · Rehab · Hearing · Technicians · Agencies · Chiropractic · Vision	163W... RN · 183500000X Pharmacist · 225200000X PTA · 2471... Radiologic Technologist	no
—	Anything this table does not carry	999900000X, or no taxonomy at all	yes — never suppressed

Every prefix verified against the live NPPES API on 2 September 2026 by reading back the code of a provider the registry itself returned. The dental grouping is the one that is not uniform, so it gets its own rule.

An unplaceable taxonomy is briefed, not suppressed. While building that table, a grouping remembered as chiropractic (16) turned out to be nursing, and chiropractic to be 11. An incomplete table costs a little noise; a wrong one silently eats a real physician’s brief. So the third outcome — *unknown* — always shows the brief.

What the rep sees instead

🚫 Not a physician — no pre-meeting brief.

NPPES NPI Registry lists TAYLOR M AAGAARD (NPI 1487300471) as Social Worker (NUCC grouping 10). Pre-meeting briefs are produced for physicians, dentists and podiatrists, so none was assembled. [Verify on NPPES NPI Registry]

Wrong person? Pick another match, or write the physician's full name on the meeting.

When the registry returns a mix, the non-doctors are simply not offered — and are still **named**: “4 further matches are not a physician (Behavior Technician, Counselor, Social Worker, Physical Therapy Assistant) and were not offered.” Silence would send a rep hunting for a different spelling.

A rep can still pick one of them from the list; the click is honoured and recorded as the meeting’s contact. What is not invented is a brief — and because the decision is stored as `not_doctor`, neither the reminder nor the meeting-body injection will produce one later, unattended.

06 The test run that shaped these rules

Six cases, run one at a time against the live system. Four fixes came out of them, listed after the table.

MEETING	RUNG	RESULT
Case obs with Dr Aaron Baas full name, no email, in the master	4	resolved Aaron P Baas · the full brief: CPT volumes, commercial signals, Lumendi account, contact intelligence, NPPES location check
Quarterly review attendee email in the master	1	resolved matched exactly, same brief
Endoscopy sync email NOT in the master	3→5	100% gate opened on the attendee's name → NICHOLAS J SHAHEEN → CMS 2024 + 2023 volumes, cross-confirmed
Meeting with Dr Katie first name only	4→5	6 × 60% none shown as the answer; the last name asked for
Dr Katie, Counselor at 200 Casentini St Salinas CA	5	100% KATIE AAB (NPI 1205257284) — first name, taxonomy and practice address all mentioned. Identified, not briefed: the registry lists her as <i>Counselor, Mental Health</i> , so §05 refuses the brief and states that instead. Every other Katie came back a counsellor, athletic trainer, PA or nurse practitioner — all named, none briefed.
Meeting with Dr Aagaard last name only	5	9 × 55% nine found, none shown, the full name asked for
... (Dentist) / ... (Behavior Technician) / ..., Counselor	5	75% Jeffrey Lynn / Charlotte Sue / Katie Diblin Aagaard respectively — one word decided each. Only the dentist is briefed; the behaviour technician and the counsellor are named and refused (§05).
Meeting with Dr Charlotte Aagaard full name	5	95% NPI 1194350884 · 4800 Sand Point Way NE, Seattle WA 98105 · phone from NPPES. A behaviour technician, so no brief (§05) — and CMS lists no Medicare services for her either, which is consistent
Review NPI 1003000126	0	100% Ardalan Enkeshafi · CMS 2024 384 services / \$44,804 · 2023 802 / \$77,382
Pipeline review	3	gate shut no name read, no lookup run

What the run changed

1. **Either half of a name is searchable — on the right field.** NPPES accepts a first name alone (verified: `first_name=KATIE&state=CA` returns the Salinas counsellor), but “Katie” was being sent as the *surname* — while the ladder had already worked out that the last name is the missing one.
2. **The gate reads the attendee’s name.** An unmatched address plus “Dr Nicholas Shaheen” on the invite was gate-blocked, so the rep got nothing at the one moment the name path exists for.
3. **Fetch wide, then trim.** Asking the registry for five candidates meant the one the meeting described could be absent from the set: the dentist is not among the first five Aagaards. Twenty are fetched, scored, and the best five offered.
4. **Nothing under 60% is offered — and the silence is explained.** The panel said “the public registries have nobody by ‘Aagaard’”. They had nine. That sends a rep looking for a different spelling instead of adding one word to the title.
5. **Only doctors are briefed.** The run made it obvious that a name search returns social workers, case managers, nurse practitioners and behaviour technicians as readily as physicians — and that briefing them was the wrong answer delivered confidently. §05 is the rule that came out of it.

07 The rules that keep it honest

Most of the work in this system is not finding data. It is refusing to state more than the data supports — because a brief is read in the two minutes before a meeting, and a rep cannot audit it.

“Could not ask” is never reported as “there is nobody”. A registry that fails to resolve produces an outage notice and a retry, not a finding about a physician. Both NPPES and CMS dropped out during testing; each time the message named the host and said, in as many words, that this is not a finding about this person.

A gap keeps its place in the layout. An outside physician renders through the same sections as a BIS one, and a field no source could fill says **“Data not available”** where it belongs. A section that quietly disappears reads as “nothing to know here” rather than “we could not find out” — Charlotte Aagaard’s brief carries fifteen such rows.

A withdrawn claim takes its consequences with it. When no candidate is confident enough to name, the NPI is dropped *and* so is everything derived from it. A facility that stands on its own evidence survives; one inferred from the discarded NPI does not.

What is extra is labelled extra. Licence numbers, taxonomy lists, Medicare participation — things the master has no column for — sit in their own block, tagged **EXTRA · not in BIS**, with the page that proves them. They are shown for context and are **not stored**.

Repetition that means something is explained. CMS lists the same CPT code twice when it was billed in both a facility and an office — commercially different things — so each row is labelled rather than deduplicated into a single misleading number.

Nothing automatic is mailed or written into a meeting. The reminder and the meeting-body injection act only on a decision the *rep* made. Both run unattended, and putting an automatic guess in front of someone in Outlook gives them no way to tell it was a guess.

08 What the rep actually gets

One renderer produces the brief, so the four places cannot disagree with each other. An outside physician gets the same block as one from the master — the same note history, the same “email me this briefing” — with three deliberate differences.

WHERE	WHEN	WHAT
The app panel	the rep opens the meeting	the brief, the shortlist, the confidence and its reasons, notes, actions
An email to themselves	on the button	the same brief, re-assembled server-side at send time so the mail and the screen cannot disagree · subject  Outside BIS: ...
The meeting’s own body	the next ingest tick	the brief written into the event in Outlook, once, organiser’s copy only — attendees are never notified
The reminder	30 minutes before	 In 27 min: Endoscopy case obs – NICHOLAS J SHAHEEN (outside BIS) with the brief and the rep’s notes

Three differences for an outside physician, each with a reason

- The brief is **injected, not fetched** — it was assembled from the sources rather than read from a table.
- The inbox-intelligence section is **skipped** — the Email Sheet is keyed to master physicians.
- “Schedule a call” is replaced by **the reason it cannot work**: NPPES publishes no email address at all, so a form that needs one is worse than an explanation.

09 Where the answer is kept

Decisions live in their own table, `outside_physician_app_meeting` — not bolted onto the meeting row and never onto the read-only master. It is **append-only**: every answer is a new row stamped with the rep it was made for and the moment it was made, and the latest row for a meeting is the effective one. A correction keeps its history rather than erasing it.

Same shape	A column for every field the app already shows for a master physician — physician, facility, contact and account fields — so both render identically. What a source could not fill stays <code>NULL</code> , and <code>NULL</code> is what the renderer turns into “Data not available”.
Who and when	<code>owner_user_id</code> , <code>owner_email</code> , <code>created_at</code> , <code>source</code> (<code>email · name · user · gate · outside</code>), <code>decided_by</code> (<code>user · system</code>), <code>confidence</code> , <code>reason</code> .
Not stored	CMS year-wise volumes, Open Payments, publications, licences. They are fetched and rendered; whether to persist them is a separate decision, deliberately not taken as a side effect of this table.
Before the table exists	The store falls back to a local SQLite file and says so, so the whole flow is testable before anybody runs the setup SQL.

10 Run it yourself

The whole ladder, from the command line — no sign-in needed

```
npm run flow:try -- "Meeting with Dr Aagaard (Dentist)"
npm run flow:try -- "Meeting with Dr Katie, Counselor at 200 CASENTINI ST SALINAS CA" --brief
npm run flow:try -- "Review NPI 1003000126" --save
npm run flow:try -- "Endoscopy sync" --email nshaheen@med.unc.edu --name "Dr Nicholas Shaheen"
```

Flags: `--city` `--state` `--taxonomy` `--address` `--zip` `--phone` supply what a meeting would otherwise carry; `--save` writes the decision; `--brief` prints the whole brief.

The real UI, on real registry answers, in a browser

```
npm run demo:page          # writes data/demo/
open data/demo/index.html  # ?open=7 jumps straight to a case
```

Nine example meetings, each captioned with the case it demonstrates. Only Graph and the session are mocked; every registry answer on the page came out of the real pipeline.

Configuration

SETTING	NOW	EFFECT
CMS_SERVICE_YEARS	2024, 2023	which Medicare years are read — more years is a config change, not a code change
REMINDER_LEAD_MINUTES	30	how early the reminder lands
INGEST_POLL_SECONDS	300	how soon a new meeting gets its brief and its injection
ANTHROPIC_API_KEY	set	unset disables every AI feature; the registry path keeps working
supabase/outside-physician-setup.sql	run in DEV	creates the decision table; without it decisions go to local SQLite

11 Open items

- **Both registries flap on this machine.** NPPES was unreachable for stretches of the test session (`ESERVFAIL`) and CMS had been the unreliable one earlier. Re-run before concluding anything about a source; `npm run enrich:doctor` diagnoses the resolver.
- **Sending a brief for real needs the rep’s Microsoft sign-in.** The message itself — subject, first line, the brief intact with its “Data not available” rows, the meeting line, the notes — is asserted in tests.
- **Taxonomy hints are read from the meeting text.** A structured field on the meeting would be stronger; today a rep types “(Dentist)” or the practice address into the title or description.
- **Outside physicians are not in the Email Intelligence Sheet**, which is keyed to master NPIs.
- **The branch is pushed; no pull request is open.** `feat-outside-bis-physician-pre-notes` , 13 commits ahead of `main` .